



A playful experience sampling method to report emotions using smartwatches and emojis

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ABSTRACT

Understanding daily emotional fluctuations has many benefits, and the Experience Sampling Method can be used to assess daily emotions. However, this method requires frequent reporting throughout the day, making continuous reporting difficult, and a simpler reporting method is required. Therefore, this study proposed a simple Experience Sampling Method using emojis and smartwatches in a playful way. To examine the usefulness of the proposed method, we conducted a 14-day survey on daily emotional fluctuations using the proposed method and the conventional text-based Experience Sampling Method and compared their reporting rates. The results showed that the proposed method had a significantly higher reporting rate than the conventional method. The new ESM method developed in this study can easily capture emotional fluctuations and is expected to be used for the evaluation and analysis of emotional states and user experiences in daily life.

1. Introduction

Mental health maintenance and improvement is a major issue for both society and science. Recently, it has become clear that understanding how emotions change in everyday life is important to maintain and improve mental health. For example, it has been shown that recording your emotional state on a daily basis increases your understanding of your own emotional state and reduces the occurrence of negative emotions in the future [1]. Notably, this method operates without the need for feedback from others and is based solely on the repeated self-recording of emotional states. Compared to interviews with counselors, recording emotions is a useful method that everyone can easily do without the assistance of a specialist. Therefore, it is possible that the development of a device that enables people to easily record their daily emotions could contribute to solving social problems related to mental health.

The Experience Sampling Method (ESM) is the most popular method for recording daily emotional fluctuations [2]. ESM is a survey method that asks respondents to report multiple times on questionnaires during a fixed period ranging from a few days to a few weeks or a few months. Previous research has shown that recording emotions using the ESM could lead to improvements in depression [3]. Therefore, recording emotions by applying the ESM may be useful for maintaining and improving mental health.

However, ESM requires participants to complete multiple surveys each day, which imposes a greater burden compared to other survey methods and makes sustained participation challenging [4]. Accordingly, this study focuses on reducing the burden of reporting by developing an ESM method that emphasizes ease of use. The type of reporting device is a critical factor in facilitating ease of response. In previous ESM studies, paper-based and smartphone-based methods have been used most frequently; however, these approaches are often time-consuming, and it has been shown that repeated reporting using such methods increases participant stress [5–7]. Therefore, employing a device that enables quick and simple responses may be essential for promoting continuous data collection. In this study, we used a smartwatch as the reporting device. Prior research has demonstrated that smartwatches, as wearable technologies, allow for faster responses than smartphones [6]. Accordingly, smartwatch-based ESM may impose less burden on participants and help facilitate the continuous recording of emotions.

The type of reporting indicator is also an important factor when considering ease of response. The most common method of reporting emotions is via text formats such as scales [8]. However, smartwatches have a small screen, and their visibility and operability are lower than those of other devices such as smartphones [9]. Therefore, when using a smartwatch to conduct ESM, it is necessary to use indicators other

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than text. For this reason, we use emojis (e.g. 😊) as the reporting indicators in this study. Kutsuzawa et al. [10] clarified that 74 facial emojis correspond to various human emotions, and they suggested the possibility of using them as emotional indicators. The advantage of using emojis is that they have a high level of visibility due to their simple design. Accordingly, the emotions they express can be interpreted more easily and intuitively than those conveyed by emotional words in text [11–13]. Taking advantage of this, emojis have recently started to be used as an indicator for measuring emotions in research such as evaluating food [14]. Therefore, emojis allow for easy recording of emotions and may be more suitable than text on small smartwatch screens.

Focusing on ease of reporting, this study proposes a new ESM method designed to minimize user burden and enable continuous recording of emotional states, and evaluates its effectiveness. Specifically, we proposed an ESM that selects emojis on a smartwatch and compare it with traditional ESM methods. In particular, this study focuses on the response rate as an evaluation indicator, comparing the response rates obtained from the ESM survey to investigate the usefulness of the developed system. Based on the above discussion, smartwatches and emojis can be reported more easily than traditional ESM methods using text and smartphones [6], so it is expected that the proposed ESM will be able to collect more responses than traditional ESM. Thus, we hypothesize that the proposed ESM will have a higher reporting rate than the traditional ESM method.

2. System

2.1. Emoji list

Kutsuzawa et al. [10] classified 74 emojis into six clusters based on their positions on two fundamental emotional axes (See Figure 1 in [10] for reference): valence (i.e., pleasant–unpleasant) and arousal (i.e., intensity of emotion). These clusters represent a wide range of human emotions. Based on this classification, we selected two emojis from each cluster to capture diverse emotional states in daily life. Especially, to account for individual differences in interpretation, we selected emojis with low variability in valence and arousal ratings (i.e., those with low standard deviations), as identified by Kutsuzawa et al. (2022). Specifically, we use the following six pair of emojis: 😡 as strong negative emotion (low valence and high arousal), 😬 as moderate negative emotion (low valence and moderate arousal), 😐 as neutral emotion near negative (moderate valence and low arousal), 😊 as neutral emotion near positive (high valence and low arousal), 😄 as moderately positive emotion (high valence and moderate arousal), and 😁 as strong positive sentiment (high valence and high arousal).

The emoji layout was designed to display all six emotional categories on a single screen, providing users with a comprehensive overview of the available emotional range without requiring navigation between multiple screens. This design choice was made to reduce cognitive load associated with scrolling and searching for appropriate emotions, enabling users to quickly identify and select the most suitable emoji for their current emotional state.

2.2. Development of an emotion reporting interface

In this study, we use Fitbit Sense as a smartwatch to implement the emotion recording system. The emotion recording system consists of two separate applications: (1) A notification application that provides nudges to encourage emotion reporting, and (2) An emotion reporting application that allows users to select emojis from multiple options. Both applications were implemented using the programming languages XML, CSS, and TypeScript, the Fitbit SDK, and Cloud Firestore provided by Google Inc. as a database. The details of each application are described below.

(1) **The nudge application** can notify the user at regular or random time intervals to prompt them to report their emotions. The screen displays the current time and an ID to identify the user, and the smartwatch vibrates at the time of the nudge. When the user receives the vibration and turns on the smartwatch screen, the application for reporting emotions (i.e., (2)) is automatically launched. With this nudge, we aimed to encourage users to always report their feelings in their daily lives.

(2) **The screen of the emotion reporting application** using the emojis is shown in Fig. 1. In this application, the user selects the emoji best suited to his or her emotions from among the 12 emojis displayed as shown in Fig. 1–(STEP1). Six emojis are displayed on one screen, and the user can switch between emojis by scrolling up and down. When the user selects an emoji, the user is taken to a confirmation screen like the one shown in Fig. 1–(STEP2). If you wish to choose an emoji again, press the BACK button at the bottom of the screen. The BACK button has a green gauge that indicates the validity time of the button. This green gauge starts moving when the user moves to the confirmation screen and runs out after three seconds. When the green gauge disappears or the screen turns off, the selected emojis, time, and other information are saved to the database on the Cloud. If the information cannot be saved in the database, it is stored in the smartwatch locally.

The design choice of implementing a three-second automatic confirmation instead of a manual confirm button was based on usability considerations identified during prototyping. This approach prevents input errors from consecutive button presses and ensures timely emotion recording by avoiding situations where users forget to manually confirm their selections, which could result in significant delays between the actual emotional state and recorded response. The three-second timer provides sufficient time for users to correct erroneous inputs using the BACK button while maintaining efficiency—users can simply leave the device untouched if their selection is correct.

3. Method

3.1. Experimental design

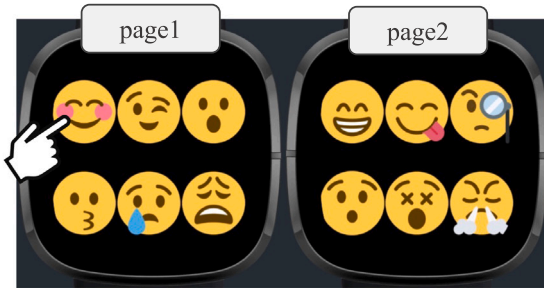
Summary of experiment. This study began with a preliminary session in which participants were briefed on the study's purpose and provided informed consent. Those who consented were then given detailed instructions on how to report their emotions according to the assigned condition. Participants were randomly assigned to either the proposed ESM group or the traditional ESM group. Groups were balanced for gender, but no other demographic variables (e.g., age, education) were controlled. Beginning the day after the briefing session, both groups completed a 14-day experience sampling method (ESM) survey, reporting their emotions 16 times per day using the assigned reporting method. The study protocol was approved by the institutional ethics committee under the protocol number BKC-LSMH-2023-058.

Participants. Participants were recruited 20 university students (10 males and 10 females, mean age 21.3 years, SD = 0.9). As a reward, participants received 10 JPY for each valid report, and an additional 1000 JPY for completing the 14-day survey. (Maximum 3240 JPY).

ESM survey. Participants in both experimental conditions were nudged 16 times per day via a designated device between 9:00 and 21:00 over a 14-day period, totaling 224 nudges. Participants were instructed to report their emotions using the designated device upon receiving a nudge. Nudges were delivered at randomly determined times, with intervals ranging from 30 to 60 min. To minimize bias in timing, random numbers were drawn from a uniform distribution, and the average interval was set to 45 min. In line with prior research suggesting

STEP1

Select an emoji appropriate to the emotion.
Scroll up and down to switch pages.

**STEP2**

Confirm the selected emoji.
Press BACK button to reselect.

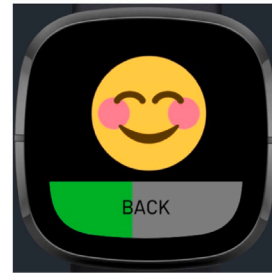


Fig. 1. Emotion reporting application using emojis.

that emotions may fluctuate over time after receiving a nudge [15], only responses submitted within 5 min were considered valid. If they were unable to respond within the 5 min window, participants were instructed to wait for the next nudge.

3.2. Emotional report in the proposed ESM

Participants assigned to the proposed ESM were instructed to wear a Fitbit Sense, which was lent to them. Fig. 2 shows the reporting procedure in this condition. At the reporting time, the smartwatch vibrated, and the emoji selection screen automatically opened. At that time, participants were instructed to select the emoji that best matched with their emotions from a list of 12 emojis. Six emojis were displayed on one screen and scrolling up and down displayed other options. After selecting the emojis, a confirmation screen was displayed, and the answer was registered three seconds later. If the user wanted to reselect the emojis, there was a BACK button at the bottom of the screen and the participant was instructed to select the emojis again. When the report was completed, (1) the user ID, (2) the selected emojis, and (3) the reporting time were recorded in the database on the cloud.

3.3. Emotional report in the traditional ESM

In a traditional ESM, nudges were sent to each participant's smartphone at a pre-set reporting time, and participants reported their emotions by open-ended description (See Fig. 3). Nudges were sent to each participant's smartphone via a Slack Bot, and participants were asked to report their emotions in Microsoft Forms. Participants were instructed to report one emotion word (e.g., sad, happy) when reporting, similar to previous studies [9]. Participants were asked to activate a vibration or sound on their smartphones to help them notice nudges from the Slack Bot. In traditional ESMs, (1) user ID, (2) one emotion word entered, and (3) reporting time were recorded.

3.4. Data analysis

To calculate the reporting rate (%), the ratio of the number of reports to the number of nudges during the 14-day experiment period was calculated (a total of 224 per participant). Next, to investigate the trends in the data collected by the proposed ESM conditions, the number of reports for each emoji reported during the survey period was calculated. Then, to investigate the trends in the data collected by the traditional ESM method, similar emotional words reported during the survey period were compiled. Three university students (three males, average age 21.7 years, SD=0.6) were asked to compile the emotion words into one emotion word based on discussions. In addition, the compiled emotion words were classified into four categories:

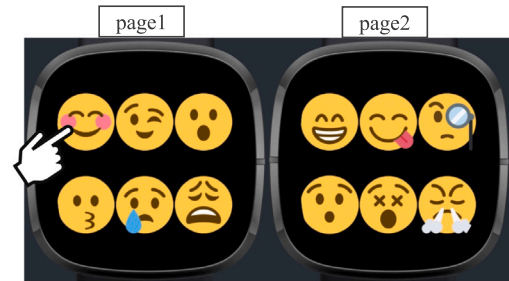
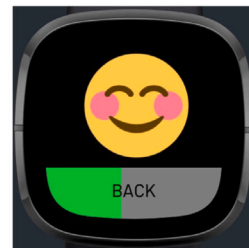
STEP1**STEP2****STEP3**

Fig. 2. Emotion reporting procedures in the proposed ESM. The procedure consists of three steps: STEP1: Participants receive vibrations and turn on their smartwatch screens, STEP2: Participants select the appropriate emoji for their emotion, and STEP3: Participants confirm the selected emoji.

“positive”, “neutral”, “negative”, and “non-emotional”. To investigate whether there is a difference in the reporting rate between the proposed ESM and the traditional ESM, we compare the reporting rates between conditions using Welch's t-test. The result of the t-test was considered statistically significant if the p values were less than 0.05, and if the effect size (Cohen's d) was greater than 0.50.

Finally, to examine whether there were differences in the distribution of emotions reported in the proposed ESM and the traditional ESM,

STEP1**STEP2**

1. 現在の感情を表す単語（悲しい、幸せなど）を入力してください *

回答を入力してください

送信

Fig. 3. Emotion reporting procedures in the conventional ESM. The procedure consists of two steps: STEP1: Participants receive notifies from Slack Bot, and STEP2: Participants input one word that describes their own emotion in the answer box.

we calculated the percentages of ‘positive’, ‘neutral’, and ‘negative’ responses by treating the total number of responses in each condition as 100%. We then used a chi-square test to compare the distributions of emotional responses between the two conditions. Results were considered statistically significant only when the p-values from the residual analysis were below 0.05.

4. Result

No participants withdrew during the ESM period, so data from all 20 participants were used in the analysis (proposed ESM: 5 males, 5 females, mean age=21.7 years, SD=0.8; traditional ESM: 5 males, 5 females, mean age=20.9 years, SD=1.0).

The total number of nudging sent during the 14-day study period for both conditions was 2240. The proposed ESM condition collected 1811 responses from 10 people (Max responses: 204, Min responses: 141, mean responses =181.1, SD=20.7). The emoji that was selected most often among the proposed ESM conditions was 😊 (405 times), followed by 😞 (246 times), 😟 (235 times), 😓 (165 times), 😊 (158 times), 😟 (149 times), 😊 (148 times), 😊 (136 times), 😟 (60 times), 😟 (60 times), 😟 (28 times), and 😟 (21 times) (see Fig. 4). An average of 11.30 (SD = 0.82) emojis were used per participant over the study period. Furthermore, an average of 5.5 (SD = 1.2) emojis were reported per participant per day.

The traditional ESM condition collected 1127 responses from 10 people (Max responses: 197, Min responses: 14, mean responses=112.7, SD=53.4). The emotions reported were compiled into 40 emotion words, with 8 positive emotion words, 10 negative emotion words, 5 neutral emotion words, and 17 non-emotional emotion words. The top five most frequently used of the 40 emotion words were “fun” for positive emotions (292 times), “sad” for negative emotions (92 times), “tired” for negative emotions (85 times), “happy” for positive emotions (85 times), and “frustrating” for neutral emotions (46 times). The number of times each word expressing an emotion was used is shown in Fig. 5.

Welch’s t-test was conducted to examine whether there was a difference between the reporting rates of the two conditions. The results showed that the proposed ESM had a significantly higher reporting rate than the traditional ESM ($t(12) = 3.77, p = 0.003, d = 1.69$). The reporting rates and standard deviations for each condition are shown in Table 1.

A chi-square test was conducted to examine whether there were differences in the distribution of emotions between the two conditions. The results revealed a significant difference in emotional distribution between conditions ($\chi^2(2) = 6.40, p = .04$). Additionally, a residual

Table 1

Response rates for the proposed ESM and the conventional ESM in the evaluation experiment.

Evaluation item	Average (Standard deviation)		<i>p</i>	Cohen’s <i>d</i>
	Proposed	Conventional		
Response rate (%)	80.8 (9.3)	50.3 (23.8)	.003	1.69

Table 2

Comparison of emotion distributions between conditions. Values indicate the percentage of each emotion category within the total responses for each condition.

Condition	Emotional category (%)		
	Positive	Neutral	Negative
Proposed ESM	46.2	28.6*	25.2
Traditional ESM	55.6	13.9*	30.4

* Denotes significant differences between conditions at the 5% level based on residual analysis.

analysis was conducted to determine which specific emotion categories contributed to this difference. The results showed that the proposed ESM yielded a significantly higher proportion of neutral emotion reports compared to the traditional ESM ($p < .05$). The distribution of emotions for each condition is shown in Table 2.

5. Discussion

5.1. Main findings

This study aimed to develop an ESM that asks users to select emojis on a smartwatch and examine its usefulness by comparing it with traditional ESM methods. We conducted a 14-days survey with 20 participants to compare the reporting rates of the proposed ESM and the traditional ESM. The results showed that the reporting rate of the proposed ESM was significantly higher than that of the traditional ESM, as we hypothesized initially.

The traditional ESM required respondents to answer the questionnaire multiple times a day, which was a burden for them and made it difficult for them to continue answering [16]. However, based on the results of this study, it was concluded that the proposed ESM is a system that can collect responses continuously more than the traditional ESM method.

One possible reason for the higher response rate for the proposed ESM than for the traditional ESM is that the reporting devices were different. Specifically, it is possible that the smartwatch was able to collect a high rate of reports because it was easier to report emotions than with a smartphone. Previous ESM surveys have shown that it is difficult to continue reporting from the perspective of the burden on respondents because it takes time to complete the report using methods such as smartphones and paper [5–7]. On the other hand, because smartwatches are wearable devices, they can report more quickly than smartphones, and it has been shown that this reduces the reporting load [6]. Therefore, it is possible that the proposal for ESM using smartwatches also resulted in a higher response rate than traditional ESM using smartphones, because it enabled quicker responses and was less burdensome.

Another reason for the higher response rate for the proposed ESM than for the traditional ESM is that the reporting indicators were different. Specifically, it is possible that the emojis were able to more easily report emotional states than text, which may have contributed to a higher reporting rate. Based on previous studies, because facial emojis are intuitive graphic tools [11,17], it has been shown that they require little cognitive effort for interpretation [11–13]. As a result, it has been found that the method can be applied to groups with low literacy skills for whom text is not available [11–13]. Therefore, it is considered that a high reporting rate was obtained because emoji

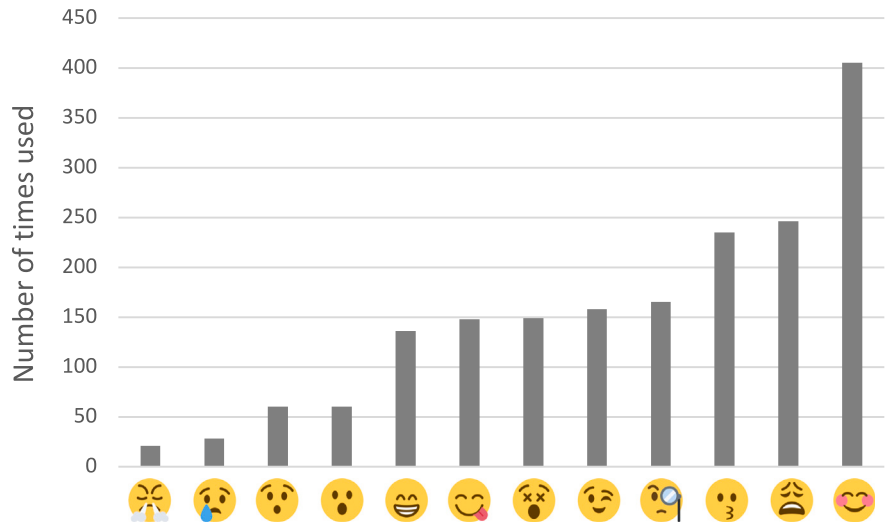


Fig. 4. Number of emojis used in the proposed ESM.

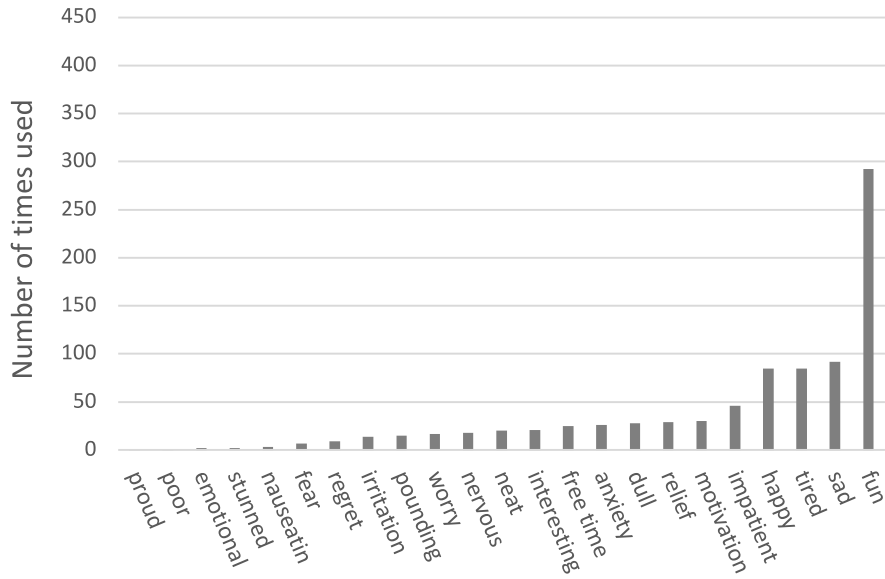


Fig. 5. Number of times emotion words are used in conventional ESM.

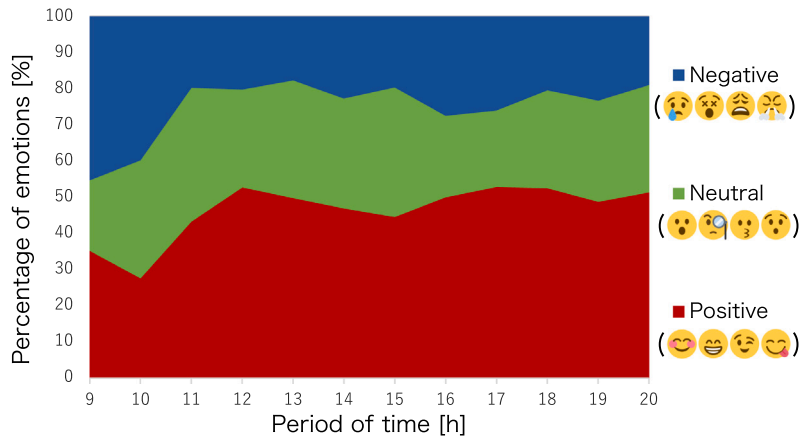


Fig. 6. Percentage of emotions per time period in the proposed ESM.

are an indicator that places less of a burden on reporting emotions than text. Based on the discussions, it can be concluded that the

high reporting rate achieved with the proposed ESM was due to a reporting device and a reporting indicator having a lower reporting

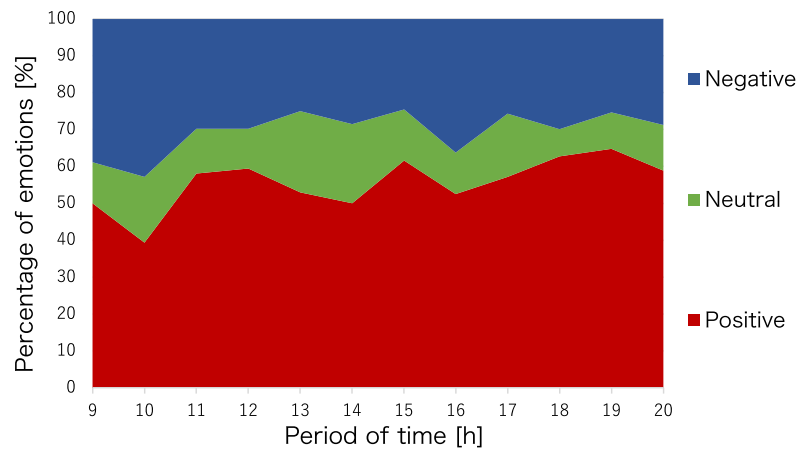


Fig. 7. Percentage of emotions per time period in the conventional ESM.

burden than those of the traditional ESM. Given the importance of emotional recording for mental health, the proposed ESM—by enabling easy and frequent emotion reporting—may contribute to its maintenance and improvement.

While the proposed ESM makes it easier to report emotions, it is possible that it may produce poorer data than the responses obtained using the traditional ESM. Specifically, because the proposed ESM only requires a simple task of choosing emojis, it is possible that a high response rate was obtained because participants did not report their emotions seriously and only chose specific emojis regardless of their emotions. Based on the results of this study, eight of the 12 emojis were selected more than 100 times during the survey period in the proposed ESM (see Fig. 4). Therefore, it is considered less likely that specific emojis were selected intensively. Moreover, no significant differences were observed in the proportion of positive and negative emotion reports between the two conditions, suggesting that the overall reporting trend was almost similar. We can see this trend in the Figs. 6 and 7. In both conditions, around half of the responses were “positive” after 10:00.

Based on these results, it seems unlikely that the proposed ESM is collecting high-quality data that is exceptionally different from that of the traditional ESM. On the other hand, the proposed ESM has more reports of neutral emotion than the traditional ESM. This difference may be caused by the ease with which emotional words are recalled. Previous studies have shown that words related to positive or negative emotions are more likely to be recalled than neutral words [18,19]. For this reason, it is possible that neutral emotions are not likely to be reported in the traditional ESM, which requires participants to recall emotions and input them as text. Therefore, the proposed ESM using emojis has the potential to capture intuitive emotions with less bias, but this point will need to be clarified through further investigation in the future.

5.2. Limitations

Device and hardware considerations

There is a possibility that only the difference in reporting devices between the proposed ESM and the traditional ESM may have affected the results. However, previous research has found no difference in the reporting rate for ESM surveys between smartphones and smartwatches [20], and it has also been pointed out that text input on smartwatches is difficult [21,22]. Therefore, even when using a smartwatch with the traditional ESM, it would be difficult to collect the same reporting rate as the proposed ESM, and it is considered that the combination of emojis and smartwatches contributed to the high reporting rate.

Comparison validity and response format

A significant limitation of this study concerns the validity of the comparison between the proposed ESM and traditional ESM methods. While we attribute the higher response rate to the combination of emojis and smartwatches, the primary difference between conditions may stem from the response format rather than the reporting medium itself. Specifically, the proposed ESM employed a multiple-choice format where participants selected from 12 predetermined emojis, whereas the traditional ESM required open-ended text input where participants freely described their emotions using words. This methodological difference introduces a confounding factor that makes it difficult to isolate whether the improved response rate resulted from (1) the intuitive nature of emojis versus text, (2) the convenience of smartwatches versus smartphones, or (3) the reduced cognitive burden of selecting from options versus generating free-form responses. Previous research has shown that multiple-choice formats generally require less cognitive effort and time compared to open-ended responses, which could partially explain the observed differences in response rates. To establish the specific contribution of emojis as reporting indicators, future studies should employ more controlled comparisons, such as comparing emoji-based versus text-based selection on the same device, or comparing emoji versus text input using identical response formats. This limitation should be considered when interpreting our findings and their implications for ESM design.

Interface design and confirmation mechanism

The three-second automatic confirmation design may have contributed to the higher response rates observed in our study, though this influence could work in multiple directions. On one hand, the automatic confirmation reduces the cognitive load and number of required actions compared to manual confirmation, potentially encouraging more frequent responses. The elimination of a final confirmation step removes a potential barrier where users might abandon the reporting process. On the other hand, the three-second delay might serve as a “cooling-off” period that encourages users to reflect on their emoji selection, potentially improving response quality. However, this brief delay could also be perceived as waiting time that might discourage some users from completing responses, particularly in time-pressured situations. Compared to the traditional ESM method, which requires users to actively submit their text responses, our automatic confirmation system reduces the number of deliberate actions required. This difference in interaction design, combined with the emoji selection format, makes it difficult to isolate the specific contribution of the confirmation mechanism to the observed response rate differences. Future studies could investigate the impact of different confirmation methods (manual button, automatic timer, immediate confirmation) within the same emoji-based system to better understand their individual effects on user engagement and response quality.

Emotional reflection and therapeutic benefits

A significant limitation of our proposed ESM system concerns the potential reduction in emotional reflection compared to traditional methods. The automatic confirmation design eliminates deliberate submission actions that might encourage users to pause and reflect on their emotional state selection. In traditional ESM, the act of typing emotion words and manually submitting responses inherently promotes deeper self-reflection, as users must actively recall, verbalize, and confirm their emotional experiences. Our emoji-based system, while achieving higher response rates, may sacrifice this reflective component that is considered essential for ESM's therapeutic benefits [1]. The ease of simply tapping an emoji and waiting for automatic confirmation, though beneficial for response rates, might not adequately promote the metacognitive awareness that previous research suggests is crucial for emotional self-monitoring benefits. A high response rate does not necessarily indicate that users are reflecting on their emotions adequately, which represents a fundamental limitation of our approach. The significantly higher proportion of neutral emotion reports in our proposed ESM (28.6% vs 13.9%) compared to traditional ESM may partly reflect this reduced reflective engagement, as users might default to neutral options when the reporting process becomes too quick or convenient. However, the three-second delay period in our system does offer an opportunity that could be leveraged to enhance participants' awareness of their emotional states. Unlike immediate confirmation systems, this waiting period could potentially encourage users to contemplate their selection, though our current study did not measure whether users actually engaged in reflection during this time. Future research should investigate whether extending this confirmation period (e.g., to 5 or 10 s) or adding explicit prompts for reflection could restore the metacognitive benefits while maintaining the convenience advantages of our emoji-based approach. This limitation must be acknowledged when considering the therapeutic applications of simplified ESM systems.

Statistical power and sample size

In this study, 10 participants for each of the proposed ESM condition and the traditional ESM condition, but the small sample size may have affected the results. Therefore, we conducted a post-hoc power analysis using the sample size and effect size obtained in this study, setting the significance level at 5%. As a result, we found that the power (i.e., $1-\beta$ error) was 94.6%. This means that the power of the analysis results is high and that the sample size of this experiment is sufficient. Therefore, it is unlikely that the small sample size affected the results.

User interface and interpretation issues

Additionally, our study did not systematically examine potential position bias effects in emoji selection, where users might preferentially choose emojis based on their screen position rather than emotional appropriateness. While our post-hoc analysis showed that emojis were selected across all positions (with the most frequently selected emoji appearing 405 times), a more rigorous investigation of position effects would strengthen our findings. Furthermore, we did not validate how users interpreted the emotional meanings of our selected emojis, relying instead on the classifications established by Kutsuzawa et al. [10]. Future studies should include user validation of emoji interpretations and systematic testing of different arrangement patterns to minimize potential position bias.

5.3. Future directions

Although this study found no significant differences between conditions in the reporting of negative emotions, emojis may nonetheless facilitate the expression of such emotions more effectively than text. Prior research [23] has shown that emojis can lower the psychological barrier to emotional expression, potentially making it easier to report emotions that are often difficult to express, particularly negative ones.

This characteristic may be especially valuable in contexts where accurately capturing negative emotional states is essential, such as mental health monitoring. Future research should further explore the potential of emoji-based reporting by refining emoji sets and user interface designs to better support the expression of negative emotion.

In this study, participants received monetary incentives to encourage participation in the survey. As a future direction, it will be important to develop a user interface (UI) that can maintain a comparable response rate even in the absence of such incentives—for example, by implementing adaptive features that nudge participants to respond at times convenient to them. Accordingly, it is necessary to collect user feedback on their experiences using the system from participants to clarify what system improvements are needed in the future.

More fundamentally, it is unclear whether the continued use of the proposed ESM is effective for maintaining and improving mental health. Previous studies have shown that continuous emotional recording promotes meta-cognition of one's own state and leads to improvements in mood [1]. Based on the results of this study, the proposed ESM has the potential to maintain and improve mental health by enabling users to easily keep a continuous record and intuitively understand their own state using emojis. Therefore, it will be necessary to investigate the effects of using the proposed ESM on daily mood in the future.

Future iterations of this system could investigate optimal emoji arrangement strategies by: (1) systematically testing different positioning patterns to minimize position bias effects, (2) conducting user validation studies to confirm emoji interpretation consistency across diverse populations, and (3) analyzing the relationship between emoji position and selection frequency using controlled experimental designs. Additionally, mapping user interpretations to standardized arousal and valence dimensions would enhance our understanding of how emoji-based emotional reporting compares to traditional text-based methods.

6. Conclusion

In this study, we developed an ESM system that allows users to report their emotions by selecting the appropriate one from several emojis displayed on a smartwatch and examined the usefulness of the developed system by comparing its reporting rate with that of traditional ESMs. The results showed that the proposed ESM showed a significantly higher reporting rate than the traditional ESM. This result may have been brought about by the combination of the smartwatch and emojis, which allowed for easier responses. The results of this study make an important contribution to solving the problem of ESM research, which is difficult to continue reporting due to the large burden on participants, from the perspective of proposing a method that can be easily reported. Furthermore, the new ESM method proposed in this study is expected to contribute to solving social problems in mental health as an emotion recording tool that can be easily used in everyday life by people who want to maintain and improve their mental health.

Beyond its immediate utility for mental health, the integration of emojis as a reporting tool presents exciting possibilities for entertainment computing. Emojis' intuitive design and universal appeal make them particularly well-suited for interactive and dynamic applications, such as games and virtual environments, where real-time emotional feedback can enhance user experience and engagement. For example, incorporating emoji-based emotion tracking into entertainment systems could enable personalized content delivery or adaptive storytelling based on user emotions. Exploring these applications could reveal new ways to leverage the playful nature of emojis to enrich both practical and entertainment-focused technologies. By demonstrating the viability of emojis in ESM, this study contributes to the growing field of entertainment computing and paves the way for innovative, emotion-aware user experiences.

CRedit authorship contribution statement

Shota Watanabe: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gaku Kutsuzawa:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yoshiyuki Kobayashi:** Writing – review & editing, Conceptualization. **Kohei Matsumura:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT and Grammarly in order to improve readability of the paper. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors have declared no conflict of interest.

Data availability

Data will be made available on request.

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